

Clarkson Institute Summer School: Examples Sheet 5 - Conjugacy and normality

Please send all comments and corrections to jmm232@cam.ac.uk.

Conjugate elements and normal subgroups

1. Starting from first principles, partition the elements of D_6 into conjugacy classes, explaining your reasoning both algebraically and geometrically. Hence, identify the normal subgroups of D_6 .

Solution: The elements of D_6 are:

$$\{e, r, r^2, r^3, r^4, r^5, m, rm, r^2m, r^3m, r^4m, r^5m\}.$$

The split into conjugacy classes is:

$$\{e\}, \quad \{r, r^5\}, \quad \{r^2, r^4\}, \quad \{r^3\}, \quad \{m, r^2m, r^4m\}, \quad \{rm, r^3m, r^5m\}.$$

This can be understood geometrically as follows:

- The identity does nothing, so is conjugate to no other symmetry.
- Rotations by $2\pi/6$ and $-2\pi/6$ (corresponding to r, r^5) must be conjugate through a reflection. They are distinct from other rotations.
- Rotations by $4\pi/6$ and $-4\pi/6$ (corresponding to r^2, r^4) must be conjugate through a reflection. They are distinct from other rotations.
- A rotation by π is conjugate to no other rotations, since it is the same clockwise as anticlockwise.
- Finally, the reflections split into two geometric classes: reflections through opposite vertices, and reflections through midpoints of opposite edges. Such reflections cannot be transformed into one another, but within each class, they can be related by rotations.

This gives all the conjugacy classes. Alternatively, we can classify the conjugacy classes algebraically using the relations between the generators:

$$\begin{aligned} (r^\alpha m^\beta)(r^{\alpha'} m^{\beta'})(r^\alpha m^\beta)^{-1} &= r^\alpha m^\beta r^{\alpha'} m^{\beta' - \beta} r^{-\alpha} \\ &= r^\alpha r^{\alpha'(-1)^\beta} m^{\beta'} r^{-\alpha} \\ &= r^\alpha r^{\alpha'(-1)^\beta} r^{-\alpha(-1)^{\beta'}} m^{\beta'}. \end{aligned}$$

Setting $\beta' = 0$, we see that a rotation $r^{\alpha'}$ is always conjugated to a rotation $r^{\alpha'(-1)^\beta}$; choosing $\beta = 0, 1$ gives $\{r^\alpha, r^{-\alpha}\}$ as a conjugacy class. On the other hand, setting $\beta' = 1$, we see that a reflection $r^{\alpha'} m$ is always conjugated to a reflection $r^{2\alpha} r^{\alpha'(-1)^\beta} m$. Taking $\beta = 0$, and varying α , we see that we can get all reflections of the form $r^{\alpha'} m, r^{\alpha'+2} m, \dots$ etc. Taking $\beta = 1$, and varying α , we see that we can get all reflections of the form $r^{-\alpha'} m, r^{-\alpha'+2} m, \dots$ etc. These are distinct conjugacy classes in D_6 , since m, r^2m, r^4m and r^5m, rm, r^3m do not overlap.

The normal subgroups are $\{e\}, \{e, r^3\}, \{e, r^2, r^4\}, \{e, r, r^2, r^3, r^4, r^5\}, \{e, r^2, r^4, m, r^2m, r^4m\}, \{e, r^2, r^4, rm, r^3m, r^5m\}$ and D_6 itself.

2. Partition the elements of Q_8 into conjugacy classes. Hence, identify the normal subgroups of Q_8 .

Solution: Observe that $\{1\}$, $\{-1\}$ constitute their own conjugacy classes, since 1 has order one, -1 has order two, and all other elements in Q_8 have order four. Recall also that an element and its inverse belong to the same conjugacy class, so $i, -i$ are in the same conjugacy class, $j, -j$ are in the same conjugacy class, and $k, -k$ are in the same conjugacy class.

These pairs could remain their own separate conjugacy classes. Alternatively, we could merge some of them. But recall that the order of a conjugacy class must divide the order of a group, so if we merged say $\{i, -i\}$ and $\{j, -j\}$, we could not also merge $\{k, -k\}$. But this would break the cyclic symmetry $i \rightarrow j \rightarrow k$ of the quaternion group. So the complete list of conjugacy classes must be:

$$\{1\}, \quad \{-1\}, \quad \{i, -i\}, \quad \{j, -j\}, \quad \{k, -k\}.$$

Using the list of subgroups from Examples Sheet 3, Question 5, we see that all the subgroups are normal.

3. Show that if $H \leq G$ is a subgroup of a group G which only has two distinct left cosets in G , then H must be normal.

Solution: Let the left cosets be H, xH . Then the right cosets must be H, Hx (since $x \notin H$). In both cases the cosets partition the group. The complement of H must be the same in both cases, which gives $xH = Hx$; hence the left and right cosets of H coincide, which is one of the conditions for normality from the lecture.

4. The *centre* $Z(G)$ of a group G is the set of all elements which lie in singleton conjugacy classes,

$$Z(G) = \{z \in G : gzg^{-1} = z \text{ for all } g \in G\}.$$

- (a) Show that $Z(G)$ is an Abelian normal subgroup of G .
- (b) Show that if H is a normal subgroup of G with $|H| = 2$, then $H \leq Z(G)$.
- (c) Determine $Z(D_4)$.

Solution:

- (a) First, suppose that $z, z' \in Z(G)$. Then we have $zz' = zz'z^{-1}z = z'z$, using the fact that z' is its own conjugate. Thus the elements of the subset $Z(G)$ all commute with one another. Further, $Z(G)$ is clearly a union of all the singleton conjugacy classes, so if it is a subgroup, it is certainly normal. It remains to verify the group axioms for $Z(G)$:
 - (G1) *Closure.* If $z, z' \in Z(G)$, then for all $g \in G$, we have $gzz'g^{-1} = gzg^{-1}gz'g^{-1} = zz'$, so $zz' \in Z(G)$ as required.
 - (G2) *Associativity.* This is inherited from the larger group G .
 - (G3) *Identity.* The identity is contained in $Z(G)$, since $geg^{-1} = gg^{-1} = e$ for all $g \in G$.
 - (G4) *Inverses.* If $z \in Z(G)$, then for all $g \in G$, we have $gzg^{-1} = z$. Taking the inverse of both sides, we see that $gz^{-1}g^{-1} = z^{-1}$, which implies that $z^{-1} \in Z(G)$, so $Z(G)$ contains all its inverses.

- (b) If $|H| = 2$, then it takes the form $\{e, h\}$ where $h^2 = e$. If H is normal, we have $\{e, h\} = g\{e, h\}g^{-1} = \{e, ghg^{-1}\}$ for all $g \in G$. This implies that $ghg^{-1} = h$, so $e, h \in Z(G)$. Thus $H \subseteq Z(G)$ as required.
- (c) $Z(D_4) = \{e, r^2\}$, since it is the union of the singleton conjugacy classes.

5. Which finite groups have exactly two conjugacy classes?

Solution: Let the group have order n . Then $\{e\}$ forms a conjugacy class, and the other conjugacy class has size $n - 1$. But the size of a conjugacy class must divide n , hence we require $n - 1$ divides n . This occurs only when $n = 2$, giving C_2 as the only finite group with exactly two conjugacy classes.

Quotient groups

6. Using your results from Questions 1 and 2, construct all possible quotients of D_6 and Q_8 . Identify standard cyclic and dihedral groups to which the quotients are isomorphic.

Solution: The quotients of D_6 are, for each subgroup H , given by:

- When $H = \{e\}$, we have $D_6/H \cong D_6$ trivially.
- When $H = \{e, r^3\}$, we have:

$$D_6/H = \{H, rH, r^2H, mH, rmH, r^2mH\} \cong D_3,$$

since $(rH)^3 = H$, $(mH)^2 = H$, and $(rH)(mH) = (mH)(rH)^{-1}$, which are the defining relations of D_3 .

- When $H = \{e, r^2, r^4\}$, we have:

$$D_6/H = \{H, rH, mH, rmH\} \cong D_2,$$

since $(rH)^2 = H$, $(mH)^2 = H$, $(rmH)^2 = I$, so all non-identity elements have order two, giving the Klein-four group.

- When $H = \{e, r, r^2, r^3, r^4, r^5\}$, we have:

$$D_6/H = \{H, mH\} \cong C_2.$$

- When $H = \{e, r^2, r^4, m, r^2m, r^4m\}$, we have:

$$D_6/H = \{H, rH\} \cong C_2.$$

- When $H = \{e, r^2, r^4, rm, r^3m, r^5m\}$, we have:

$$D_6/H = \{H, rH\} \cong C_2.$$

- Finally, when $H = D_6$, we obviously have $D_6/D_6 \cong C_1$.

We can perform a similar classification with the quaternion group. The quotients of Q_8 are, for each subgroup H , given by:

- When $H = \{1\}$, we obviously have $Q_8/H \cong Q_8$.
- When $H = \{1, -1\}$, we have $Q_8/H = \{H, iH, jH, kH\} \cong D_2$, since each non-identity element has order two.
- When $H = \{1, -1, i, -i\}$, we have $Q_8/H = \{H, jH\} \cong C_2$. Similarly when $H = \{1, -1, j, -j\}$ and $H = \{1, -1, k, -k\}$.
- When $H = Q_8$, we trivially have $Q_8/Q_8 \cong C_1$.

7. Let m be a reflection in D_4 . Show that $\{e, m\}$ is not a normal subgroup. Construct its left cosets, and hence show that coset multiplication is not well-defined in this instance.

Solution: Observe that $r\{e, m\}r^{-1} = \{e, rmr^{-1}\} = \{e, r^2m\}$, so this subgroup does not contain all of its conjugates. It is therefore not normal. Let the subgroup be $H = \{e, m\}$; then the left cosets are given by:

$$H = \{e, m\}, \quad rH = \{r, rm\}, \quad r^2H = \{r^2, r^2m\}, \quad r^3H = \{r^3, r^3m\}.$$

Computing the product of cosets, we have:

$$H(rH) = \{e, m\} \cdot \{r, rm\} = \{r, rm, mr, mrm\} = \{r, rm, r^3m, r^3\},$$

which is not a left coset. Hence the coset multiplication is not well-defined in this case.

8. In theoretical physics, a *Lorentz boost* is a symmetry transformation of $(1+1)$ -dimensional spacetime represented by the matrix:

$$\begin{pmatrix} \cosh(\phi) & -\sinh(\phi) \\ -\sinh(\phi) & \cosh(\phi) \end{pmatrix},$$

where ϕ is a real parameter, called the *rapidity* of the Lorentz boost. It represents a transformation between a stationary frame of reference, with coordinates (x, ct) , and a moving frame of reference, (x', ct') .

- Show that the set H of Lorentz boosts forms a group.
- The set of Lorentz boosts is now extended with time-reversal symmetry T , and parity inversion P , represented by the matrices:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad P = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Let G be the group generated by the elements of H , together with T and P . Show that $H \trianglelefteq G$ is a normal subgroup of G . Is the group K , generated by T and P only, a normal subgroup of G ?

- Construct the quotient group G/H , and identify it with a standard group.

Solution:

(a) Verifying each of the axioms in turn, we have:

(G1) *Closure*. Multiplying two matrices of the given form, we have:

$$\begin{aligned} & \begin{pmatrix} \cosh(\phi) & -\sinh(\phi) \\ -\sinh(\phi) & \cosh(\phi) \end{pmatrix} \begin{pmatrix} \cosh(\theta) & -\sinh(\theta) \\ -\sinh(\theta) & \cosh(\theta) \end{pmatrix} \\ &= \begin{pmatrix} \cosh(\phi)\cosh(\theta) + \sinh(\phi)\sinh(\theta) & -\cosh(\phi)\sinh(\theta) - \sinh(\phi)\cosh(\theta) \\ -\sinh(\phi)\cosh(\theta) - \sinh(\theta)\cosh(\phi) & \sinh(\phi)\sinh(\theta) + \cosh(\phi)\cosh(\theta) \end{pmatrix} \\ &= \begin{pmatrix} \cosh(\phi + \theta) & -\sinh(\phi + \theta) \\ -\sinh(\phi + \theta) & \cosh(\phi + \theta) \end{pmatrix}. \end{aligned}$$

Thus the set is closed under matrix multiplication.

(G2) *Associativity*. Matrix multiplication is associative.

(G3) *Identity*. Taking $\phi = 0$, we see that the identity I is contained in the group.

(G4) *Inverses*. The inverse of the matrix with parameter ϕ is the matrix with parameter $-\phi$, as we can see from the matrix multiplication law.

(b) Define the Lorentz boost matrix with parameter ϕ as $B(\phi)$. We observe that $TP = PT$, that:

$$\begin{aligned} TB(\phi) &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cosh(\phi) & -\sinh(\phi) \\ -\sinh(\phi) & \cosh(\phi) \end{pmatrix} \\ &= \begin{pmatrix} \cosh(\phi) & -\sinh(\phi) \\ \sinh(\phi) & -\cosh(\phi) \end{pmatrix} \\ &= \begin{pmatrix} \cosh(\phi) & \sinh(\phi) \\ \sinh(\phi) & \cosh(\phi) \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= B(-\phi)T. \end{aligned}$$

Similarly, we have $PB(\phi) = B(-\phi)P$. In particular, any product of powers of $T, P, B(\phi)$ can be arranged using these relations so that it takes the form:

$$B(\phi)T^\alpha P^\beta.$$

Now, to show that the subgroup of Lorentz boosts is normal, we note that:

$$\begin{aligned} (B(\phi)T^\alpha P^\beta)B(\phi')(B(\phi)T^\alpha P^\beta)^{-1} &= B(\phi)T^\alpha P^\beta B(\phi')P^{-\beta}T^{-\alpha}B(-\phi) \\ &= B(\phi)T^\alpha B((-1)^\beta \phi')T^{-\alpha}B(-\phi) \\ &= B(\phi)B((-1)^{\alpha+\beta} \phi')B(-\phi) \\ &= B((-1)^{\alpha+\beta} \phi'). \end{aligned}$$

This is a Lorentz boost, so indeed H is normal in G . To see whether the subgroup K is normal, we calculate:

$$\begin{aligned}
 (B(\phi)T^\alpha P^\beta)T^{\alpha'}P^{\beta'}(B(\phi)T^\alpha P^\beta)^{-1} &= B(\phi)T^\alpha P^\beta T^{\alpha'}P^{\beta'}P^{-\beta}T^{-\alpha}B(-\phi) \\
 &= B(\phi)T^{\alpha'}P^{\beta'}B(-\phi) \\
 &= T^{\alpha'}B((-1)^{\alpha'}\phi)P^{\beta'}B(-\phi) \\
 &= T^{\alpha'}P^{\beta'}B((-1)^{\alpha'+\beta'}\phi)B(-\phi).
 \end{aligned}$$

Hence we see that taking $\alpha' = 1, \beta' = 0$, we have an element that is not of the form $T^\alpha P^\beta$, so the subgroup is *not* normal.

(c) The quotient G/H consists of the cosets:

$$H, \quad TH, \quad PH, \quad TPH.$$

Since $T^2 = P^2 = (TP)^2 = I$, this group is just D_2 .

9. By naturally identifying C_n with a subgroup of C_{nm} , prove that:

$$\frac{C_{nm}}{C_n} \cong C_m.$$

Deduce the structure of all possible quotients of C_{24} .

Solution: We can write:

$$C_{nm} = \{e, r, r^2, \dots, r^{nm-1}\}.$$

The subgroup that we should identify with C_n is given by:

$$C_n = \{e, r^m, r^{2m}, \dots, r^{m(n-1)}\}.$$

Consequently, the cosets of C_n are given by:

$$C_{nm}/C_n = \{C_n, rC_n, r^2C_n, \dots, r^{m-1}C_n\}.$$

Since $(rC_n)^m = C_n$, we see that this is a cyclic group of order m , giving the isomorphism in the question.

For the given group C_{24} , we have factors 1, 2, 3, 4, 6, 8, 12, 24. Thus all possible quotients are:

$$\begin{aligned}
 C_{24}/C_{24} &\cong C_1, & C_{24}/C_{12} &\cong C_2, & C_{24}/C_8 &\cong C_3, & C_{24}/C_6 &\cong C_4, \\
 C_{24}/C_4 &\cong C_6, & C_{24}/C_3 &\cong C_8, & C_{24}/C_2 &\cong C_{12}, & C_{24}/C_1 &\cong C_{24}.
 \end{aligned}$$

10. Show that both H and K may be naturally identified with normal subgroups of $H \times K$. Hence, prove that:

$$\frac{H \times K}{H} \cong K \quad \text{and} \quad \frac{H \times K}{K} \cong H.$$

Solution: We saw in the proof of the direct product theorem that we should identify H with elements of the form (h, e) in $H \times K$, and K with elements of the form (e, k) in $H \times K$. This implies that:

$$\frac{H \times K}{H} = \{(e, k)H : k \in K\} \cong K, \quad \frac{H \times K}{K} = \{(h, e)K : h \in H\} \cong H,$$

trivially.

11. Let G be a group. Show that $G/Z(G)$ is cyclic if and only if G is Abelian.

Solution: It is useful to note that $Z(G)$, the set of all elements with singleton conjugacy classes, is also the set of elements which commute with all other elements. This is because if $z \in Z(G)$, we have $gzg^{-1} = z$ for all $g \in G$, giving $gz = zg$.

If G is Abelian, then $Z(G) = G$, because all elements lie in singleton conjugacy classes. On the other hand, if $G/Z(G)$ is cyclic, then the cosets all take the form:

$$(xZ(G))^k = x^k Z(G)$$

for some element $x \in G$. But now given any two elements of the group G , we may express them in the form $x^k z, x^l z'$ for some $z, z' \in Z(G)$. In this form, it is clear that they commute, since z, z' commute with everything. Therefore G is Abelian.

Common misconceptions about quotients

12. Is it true that for any group G with a normal subgroup $H \trianglelefteq G$, we must have:

$$\frac{G}{H} \times H \cong G?$$

Give a proof or counterexample as appropriate.

Solution: This is false - a basic counterexample uses $D_3 = \{e, r, r^2, m, rm, r^2m\}$. Observe that $H = \{e, r, r^2\}$ is a normal subgroup. The quotient is given by:

$$D_3/H = \{H, mH\} \cong C_2.$$

But it is not true that $D_3 \cong D_3/H \times C_2$; we saw on Examples Sheet 4, Question 2, that D_3 is not a direct product of subgroups.

13. If $K \trianglelefteq H$, and $H \trianglelefteq G$, is it true that $K \trianglelefteq G$? Provide a proof or counterexample as appropriate.

Solution: This is false; a counterexample comes from the chiral tetrahedral group $G = T$. Let $H = \{I, R, G, B\}$ be the subgroup of rotations by π about the axes through midpoints of opposite edges, together with the identity. This is certainly normal, because conjugates of these rotations cannot produce rotations by $2\pi/3$ or $4\pi/3$, the rotations of the faces.

Further, $H = \{I, R, G, B\}$ is a group of order four, so is Abelian. Thus all subgroups are normal. This implies that $K = \{I, R\} \trianglelefteq H$ is a normal subgroup. Hence we have $K \trianglelefteq H$, $H \trianglelefteq G$. But K is not normal in G , because the conjugates of R are not all contained in K (we can conjugate any of the π rotations into any of the other π rotations, using a $2\pi/3$ or $4\pi/3$ rotation).

14. Let $H, K \trianglelefteq G$ be normal subgroups of a group G with $H \cong K$. Is it true that $G/H \cong G/K$? Provide a proof or counterexample as appropriate.

Solution: Once again, this is false. Consider $G = C_4 \times C_2$, with $C_4 = \{e, r, r^2, r^3\}$ and $C_2 = \{e, s\}$. We note that all subgroups are normal, since the group is Abelian. Further, we note that $H = \{(e, e), (r^2, e)\} \trianglelefteq C_4 \times C_2$, and $K = \{(e, e), (e, s)\} \trianglelefteq C_4 \times C_2$ are both subgroups, isomorphic to C_2 in both cases. However, the quotients are given by:

- $G/H = \{H, (r, e)H, (e, s)H, (r, s)H\}$, which is isomorphic to D_2 , since $((r, e)H)^2 = (r^2, e)H = H$, $((e, s)H)^2 = (e, e)H = H$ and $((r, s)H)^2 = (r^2, e)H = H$.
- $G/K = \{K, (r, e)K, (r^2, e)K, (r^3, e)K\}$, which is isomorphic to C_4 .

Since $D_2 \not\cong C_4$, we have $G/H \not\cong G/K$, even though $H \cong K$.

Bonus questions

15. (*) Let G be a finite non-Abelian group.

- (a) Explain why the size of $Z(G)$ can be at most a quarter the size of G .
- (b) Hence, show that the chance of picking two elements from G at random that commute is $5/8$. Show that this bound cannot be improved.

Solution:

- (a) Since G is finite, the order of $Z(G)$ must divide the order of G . If $|G|/|Z(G)| = 1$, then the group is Abelian. If $|G|/|Z(G)| = 2$, then the quotient $G/Z(G)$ is isomorphic to C_2 , and hence cyclic. Thus the group is Abelian. Similarly if $|G|/|Z(G)| = 3$. Thus the size of $Z(G)$ can be at most a quarter the size of G in a non-Abelian group.

- (b) Pick an element g at random from G . The chance that it is in the centre is at most $1/4$. If we then select another element h from G , and g was in the centre, they automatically commute. Otherwise, g is not in the centre with chance at least $3/4$. Then the group $C(g)$ of elements that leave g invariant under conjugation (from the lecture) is precisely the set of elements that commute with g . This can have size at most $1/2$ of G , since it cannot be all of G and must have size a factor of the size of G . Thus the chance that we subsequently draw h from $C(g)$ is at most $1/2$.

The overall chance of drawing two commuting elements is therefore at most $1/4 + (3/4) \cdot (1/2) = 5/8$, as required.

To show that the bound cannot be improved, we must exhibit a group saturating the bound. An example saturating the bound is the quaternions. Examining the group table from the lecture notes, there are 40 pairs that commute out of a total of 64 possible pairs, giving a fraction of $40/64 = 5/8$ commuting pairs of elements.